

Beyond Cache Emulation: Advantage-Preserving Large-to-Small KV State Handoff without Re-Prefill

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Abstract

Large language model systems increasingly rely on model routing, mid-session switching, and multi-agent collaboration, yet a receiver typically has to prefill the full context again after a model switch. Prior work separately establishes the feasibility of cross-model KV replacement and semantic communication through latent/KV states, but leaves a stronger question open: can a weaker student that never constructs context self-KV consume only a stronger teacher’s runtime state and inherit part of the teacher’s task advantage? We study Advantage-Preserving Large-to-Small KV Handoff and introduce APCS, which separates state synthesis into a student-compatible base and an advantage-carrying state. Before offline experiments, we use a conservative prior projection (not measured evidence) to make the hypothesis quantitatively falsifiable, and we report it as ranges rather than fabricated point estimates so that a pre-registered hypothesis is not mistaken for a measured result. For Qwen3-4B→1.7B at 4K, we project that a ridge replacement retains roughly 92–96% of the student self-prefill score while still showing CHG in the -2 to -4 pp band; a low-rank compatibility base retains roughly 90–94% at $\text{PCR} \approx 0.23$; and full APCS plausibly lands in a CHG range of about -1.5 to $+3$ pp (TGRR roughly -15% to $+30\%$), a range whose central tendency is only mildly positive and does not exclude zero, while retaining on the order of 8–16% student-side prefill saving. A single-GPU cost projection suggests little or negative benefit at very short contexts, with the map/load-versus-prefill break-even plausibly falling somewhere in the 4K–8K range rather than at a single fixed point, since part of the mapping cost may itself scale with context length. These values are experimental priors only, deliberately left wide to reflect genuine pre-experiment uncertainty; every quantitative claim in the submission will be replaced by frozen-test, multi-seed, measured P50/P95 results.

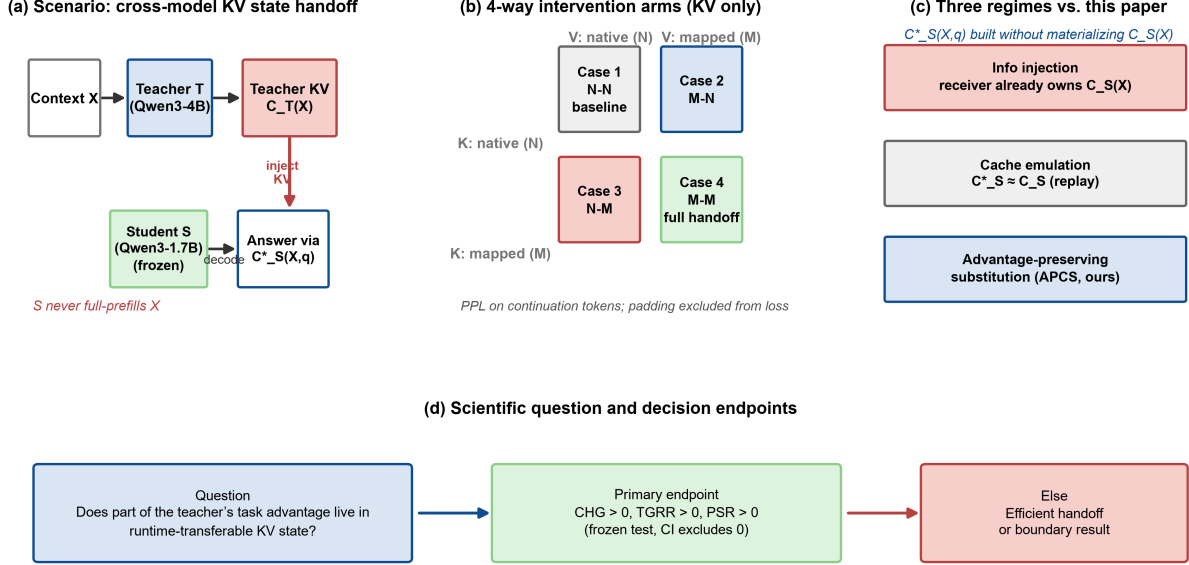
1 Introduction

When one model finishes reading a long context and a second, smaller model takes over, the second model normally has to re-read the whole context. Prefill is already the dominant cost of long-context inference, and multi-model serving makes that cost recur: routing, mid-session switching, and multi-agent setups pass the same context between models, and each receiver pays the prefill again.

Existing systems reuse a cache only within one model, or they impose architectural constraints. Droid-Speak shares KV state across fine-tuned variants of one architecture, ICarus and PrefillShare redesign the model so several decoders share a prefill module, and Cross-Model KV Cache Transfer learns closed-form ridge or MLP mappings that let a receiver skip prefill for matched-KV pairs inside one family. A second line of work treats states as a communication medium: Cache-to-Cache fuses source KV into the receiver’s own state, LatentAlign aligns KV latent spaces, and Latent Cache Flow compresses cache-level communication. None of these asks whether the receiver can do better than its own full-prefill baseline.

We ask that question directly: does part of the task advantage of a stronger teacher live in the runtime KV state, in a form a weaker student can consume without re-reading the original context and without changing the student’s weights? A positive answer would turn cross-model KV handoff from redundant-compute elimination into runtime capability transfer. A negative answer would draw a clean boundary between state compatibility and capability transfer. Both outcomes are useful, which is why the paper is organized around a falsifiable probe rather than a promised gain.

The paper makes three contributions: a problem definition (advantage-preserving large-to-small KV handoff, with CHG and PSR as the joint criterion), a method that separates a student-compatible base state from an advantage-carrying state (APCS), and an experimental characterization plan that treats teacher gap,



Evidence gates: state compatibility $\neq$ capability transfer; representation similarity is not a substitute for task-level CHG.

Figure 1: Problem setup and research question. (a) The teacher reads a long context and produces KV state; a mapper synthesizes a student-compatible cache $C^*_S(X, q) = F_\theta(C_T(X), q)$, which is injected into the frozen student without full prefill. (b) The four diagnostic arms switch native and mapped K/V. (c) How this setting differs from information injection and cache emulation. (d) The decision rule: capability transfer if CHG, TGRR, and PSR are positive on frozen tests, otherwise Efficient State Handoff or a boundary result.

model-size gap, context length, and mapper budget as variables and reports direction asymmetry rather than hiding it.

2 Related Work

2.1 KV Cache and the Economics of Prefill

A Transformer prefill pass processes all input tokens in parallel and stores layer-wise keys and values for subsequent autoregressive decoding. FlashAttention and FlashAttention-2 substantially improve attention IO efficiency, but they do not remove redundant prefill when a second model must reprocess the same long context [2, 3]. PagedAttention/vLLM reduces KV fragmentation and enables intra-model cache sharing, while Mooncake and LMCache elevate KV state into a transferable storage object across GPU, CPU, SSD, and inference engines [4, 5, 6]. These systems motivate our cost accounting but predominantly assume state reuse within the same model.

2.2 Cross-Model KV Reuse and Prefill Replacement

DroidSpeak targets fine-tuned model variants with the same architecture and selectively recomputes a small set of critical layers while reusing the remaining KV state. ICarus conceptually factorizes a decoder-only Transformer into a logical encoder that produces KV states and a logical decoder that predicts tokens, enabling multiple specialized decoders to share an identical cache. PrefillShare similarly introduces a shared prefill module with specialized decoding modules. These works establish cross-model redundant prefill as a systems problem, but they impose architectural or training-time constraints.

The closest replacement baseline is Cross-Model KV Cache Transfer [7]. For within-family pairs with matched KV-head count and per-head dimension, it selects top- k predictive source layers, removes RoPE

from keys before fitting, and learns per-head ridge mappings from source to receiver KV. Its results directly establish the feasibility of receiver-free cross-size state replacement, so neither receiver-free inference nor a linear mapper is a novelty claim of this work. Our optimization target is different: rather than projecting teacher state back toward the student’s self-prefill state as faithfully as possible, we ask whether a receiver-consumable state can preserve useful teacher-only task advantage.

2.3 KV/Latent States as an Inter-Model Communication Medium

C2C trains a neural projector that fuses a source model’s KV cache into the target model’s own KV cache and gates the layers that benefit from communication [8]. LatentAlign maps multiple models into and out of a shared KV latent space [9]. LCF compresses cache-level communication and transmits information absent from the receiver context [10]. KaVa further shows that compressed teacher KV trajectories can act as supervision for latent-reasoning students [14]. These works support the premise that runtime states contain semantically useful signals, but they do not answer our strict substitution setting because the receiver typically has its own context-derived state or the transfer occurs during training rather than runtime handoff.

2.4 Behavioral Stability and Long-Context Evaluation

Numerical KV similarity alone is an incomplete fidelity criterion. Heo et al. report that attention-output cosine is more predictive of downstream retention than raw KV R^2 [7]. Liang et al. show an even stronger failure mode: KV reuse can preserve end-task accuracy while substantially changing an LLM judge’s candidate-selection behavior, motivating Judge Consistency Rate and cross-candidate attention diagnostics [15]. We therefore evaluate representation-, attention-, behavior-, and task-level fidelity. For long contexts, LongBench, RULER, and LongBench Pro provide complementary realistic and controllable evaluation regimes [17, 18, 19].

3 Problem Formulation

3.1 Setting

Let X denote a long context, T a teacher model, S a student model, and q a later query. The teacher has already processed X as part of an upstream reading, reasoning, or conversational stage and materialized its runtime KV state $C_T(X)$. The student is not allowed to full-prefill X and instead consumes a synthesized student-compatible cache:

$$C_S^*(X, q) = F_\theta(C_T(X), q), \quad (1)$$

$$\hat{y} = S(q; C_S^*(X, q)), \quad \text{Prefill}_S(X) = 0. \quad (2)$$

3.2 State Substitution versus Information Injection

We distinguish three regimes. Information injection assumes that the receiver already owns $C_S(X)$ and augments it with teacher information. Cache emulation constructs a state from $C_T(X)$ that approximates the behavior of $C_S(X)$. Our target, advantage-preserving state substitution, constructs C_S^* without ever materializing $C_S(X)$ and asks whether this substituted state can outperform the student self-prefill baseline. This distinction defines our boundary with C2C/LCF and with conventional cross-model cache emulation.

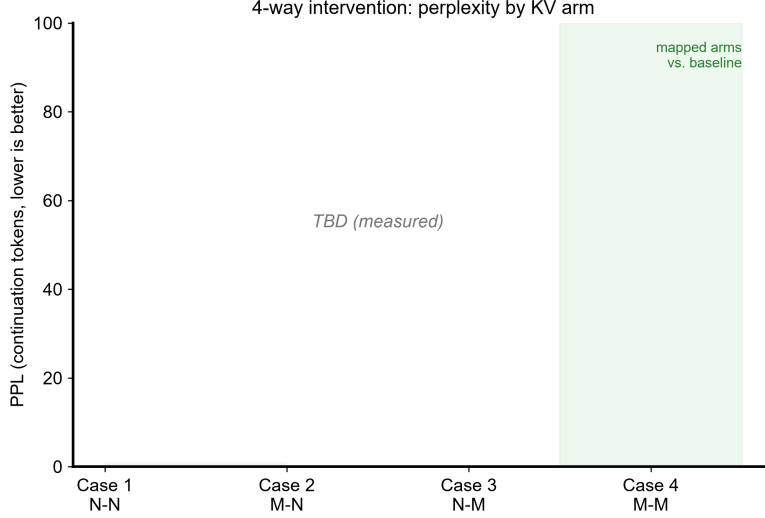


Figure 2: The four-way intervention design. Each arm is a different assignment of native or ridge-mapped K and V to the student’s `past_key_values`. PPL is computed on continuation tokens with padding excluded from the loss; the mapped arms are highlighted because their gap to the native baseline is the quantity of interest.

3.3 Metrics and Pareto Criterion

$$\text{CHG} = \text{Score}_{\text{handoff}} - \text{Score}_{\text{student-self}}, \quad (3)$$

$$\text{TGRR} = \frac{\text{Score}_{\text{handoff}} - \text{Score}_{\text{student}}}{\text{Score}_{\text{teacher}} - \text{Score}_{\text{student}}}, \quad (4)$$

$$\text{PSR} = 1 - \frac{T_{\text{map}} + T_{\text{load}} + T_{\text{query}}}{T_{\text{prefill}, S(X)}}, \quad (5)$$

$$\text{PCR} = \frac{\text{MapperSize}_{\text{ours}}}{\text{MapperSize}_{\text{ridge}}}. \quad (6)$$

CHG is the primary scientific endpoint: it asks whether the handoff student exceeds its own self-prefill baseline. TGRR measures the fraction of the original teacher–student performance gap recovered by handoff. PSR measures student-side prefill saving only in the main scenario where the teacher cache already exists for an upstream reason. PCR constrains deployment footprint relative to the cross-model ridge mapper. We call $\text{CHG} > 0$ together with $\text{PSR} > 0$ a Pareto Capability Handoff.

4 Method: Advantage-Preserving Cache Synthesis

4.1 Design Principle

APCS starts from a tension between compatibility and advantage preservation. A mapper trained only to minimize numerical error to the student self-KV may become very good at reproducing what the student would have computed itself, while actively discarding teacher-specific state directions that could improve the downstream task. We therefore separate two objectives: first construct a stable receiver-consumable compatibility state; then learn a lightweight advantage state that is not fully constrained to collapse back into the student self-prefill manifold.

4.2 Compatibility Base

$$C_{\text{base}}(X) = F_{\text{compat}}(C_T(X)). \quad (7)$$

F_{compat} begins from the strongest directly comparable replacement baseline. For each student layer/head, we select top- k teacher source layers; keys are mapped in de-RoPE content space and then re-encoded with the student positional scheme; K and V are mapped separately. The minimal-resource study first reproduces ridge regression and then explores low-rank factorization, shared bases, and head/layer parameter sharing to reduce PCR while maintaining retention.

4.3 Advantage State

$$R_{\text{adv}}(X) = E_{\text{adv}}(C_T(X)). \quad (8)$$

Once the compatibility base is stable, APCS adds a lightweight advantage state R_{adv} . Unlike the base mapper, this branch is not forced to reconstruct the student self-KV; it is supervised to retain teacher-conditioned directions that improve the downstream task while remaining consumable by the frozen student. The minimal configuration uses a global or layer-wise scalar α :

$$C_S^*(X) = C_{\text{base}}(X) + \alpha \cdot R_{\text{adv}}(X). \quad (9)$$

A full configuration may later introduce lightweight query adaptation $g(q)$, but it must not rescan the long teacher cache for each new query. All $O(|X|)$ context synthesis should happen once; per-query adaptation should operate only over short-query features and layer/head or small latent-slot gates.

4.4 Training Objective

$$\mathcal{L}_{\text{APCS}} = \mathcal{L}_{\text{task}} + \lambda \mathcal{L}_{\text{self}} + \lambda \mathcal{L}_{\text{teacher}} + \lambda \mathcal{L}_{\text{att}} + \lambda \mathcal{L}_{\text{reg}}. \quad (10)$$

$\mathcal{L}_{\text{task}}$ directly optimizes the task answer or suffix generation objective. $\mathcal{L}_{\text{self}}$ stabilizes student behavior on examples where the teacher has little advantage or may be wrong. $\mathcal{L}_{\text{teacher}}$ provides teacher guidance only when a teacher advantage is established on the training split. $\mathcal{L}_{\text{att}}$ aligns downstream-relevant attention outputs rather than only raw KV values. $\mathcal{L}_{\text{reg}}$ controls rank, sparsity, or mapper budget. Teacher-advantage weights are computed exclusively from training data; validation selects thresholds and hyperparameters; the test split remains frozen.

4.5 Complexity and Deployment Accounting

Our primary deployment scenario assumes the teacher cache already exists because the teacher processed X upstream, so the student-side handoff cost is map + cache load + query-only prefill. If the teacher is invoked solely to create a transferable cache, teacher prefill must be included and the reuse break-even must be reported.

5 Experimental Setup

5.1 Research Phases and Models

We begin with a minimal-resource probe: Qwen3-4B as teacher, Qwen3-1.7B as student, one NVIDIA GPU. Qwen3-4B needs roughly 8GB in FP16/BF16 and Qwen3-1.7B roughly 3.5GB. With the sequential pipeline of Section 5.5 (load the teacher, extract and save its KV cache, unload it, then load the student), peak GPU memory stays below 12GB. The pair stays inside one family, which keeps tokenizer, KV-head count, head dimension (128), and RoPE base compatible while preserving a real size gap: 36 layers versus 28 [16].

5.2 Tasks and Data Splits

The Fidelity Set measures whether handoff preserves basic student behavior and may include HellaSwag, ARC-Challenge, selected MMLU tasks, and WinoGrande. The Teacher-Advantage Set is selected only at the dataset level when the teacher reliably exceeds the student; individual test examples are never filtered by teacher-win/student-loss outcomes. Long-context evaluation uses LongBench/RULER-style tasks, starting at 4K before moving to 8K/16K/32K. A behavior-sensitive set contains judge/ranker or multi-candidate comparison tasks to reveal decision drift that is not visible in aggregate accuracy.

5.3 Baselines

- B0 Student Full Prefill: quality, behavior, and latency reference.
- B1 Teacher Full Inference: performance upper reference and teacher-gap definition.
- B2 Teacher Text/Summary Handoff → Student: conventional text-mediated communication.
- B3 Cross-Model Ridge Mapper: strongest directly comparable replacement baseline.
- B4 Cross-Model MLP: nonlinear repair baseline for difficult pairs.
- B5 C2C-style Fusion: capability-enhancement reference with a different receiver-state assumption.
- B6 LatentAlign / LCF: shared-latent and lightweight cache-communication references when code/checkpoints permit a fair comparison.
- O1 APCS Base Only; O2 Base + Advantage State; O3 full APCS if lightweight query adaptation is enabled.

5.4 Metrics and Statistical Protocol

All learned mappers/adapters use at least three random seeds. CHG confidence intervals and significance are estimated with paired bootstrap or paired permutation tests whenever applicable. Latency is measured after warm-up with repeated synchronized runs and reported at P50/P95, separately accounting for mapping, CPU-to-GPU cache load, query-only prefill, decode, and student full-prefill baseline. Every run fixes model commit, tokenizer, attention implementation, dtype, RoPE configuration, cache layout, hardware, batch size, and context length.

5.5 Minimal-Resource Implementation

The study uses Hugging Face Transformers and controls `past_key_values` directly. One GPU run follows this order: load teacher, forward and capture, move the needed state to CPU, unload teacher, load student, map and inject, decode and evaluate. The two models are never resident at the same time, so the peak stays below 12GB. Compute is small: the ridge fit uses head dimension 128 and the closed-form solution $(X^\top X + \lambda I)^{-1} X^\top Y$, so 100 calibration examples of length 2048 fit in seconds; the 100-sample forward pass on the student takes minutes; and the whole stage, debugging included, stays under half a day. Calibration uses 100 examples at 2048 tokens. The reference implementation and the three engineering details that keep the probe valid are in Appendix A.

6 Expected Outcomes and Testable Predictions

This section fixes expectations before the offline experiments and prevents post-hoc target shifting. The numbers below are prior projections, not measurements, and must not be used as submission claims. The projection assumes: ridge replacement mostly restores student behavior rather than capability; any transferable teacher advantage is modest, on the order of 1–3 percentage points in this first small pair; and on a single GPU with sequential residency, map and load overhead hurts short contexts and is amortized only as context grows.

6.1 Reporting Convention: Ranges Instead of Fabricated Precision

An earlier draft reported prior projections as single-decimal point estimates with seed-style error bars and smooth curves, borrowing the visual grammar of measured multi-seed results for numbers that were never measured. That presentation risks being read as evidence, so every prior projection below is a range with an explicit central tendency, allowed to straddle zero when the underlying hypothesis is genuinely uncertain. Every figure in this section shows a band, not a line.

6.2 Replacement Fidelity

The prior projection deliberately separates replacement from capability transfer. Full ridge is expected to reach roughly 92–96% retention while still underperforming student self-prefill; compressing the mapper to

Table 1: Prior-projected main results (NOT measured), reported as plausible ranges rather than point estimates with fabricated seed variance. Values assume Qwen3-4B→1.7B, 4K context, and a frozen student. Student/Teacher are illustrative common composite task scores on a 0–100 scale.

Method	Ctx	Student	Teacher	Handoff	Retention	CHG	TGRR	PSR	PCR
Student Full Prefill	4K	54.8	64.7	–	100.0%	0.0	0.0%	0.0%	–
Cross-Model Ridge	4K	54.8	64.7	50.4–52.6	92–96%	–4.4 to –2.2	–44 to –22%	3–8%	1.00
Low-rank Base	4K	54.8	64.7	49.3–51.5	90–94%	–5.5 to –3.3	–56 to –33%	12–20%	0.23
APCS	4K	54.8	64.7	53.2–57.5	97–105%	–1.6 to +2.7	–16 to +27%	8–16%	0.27

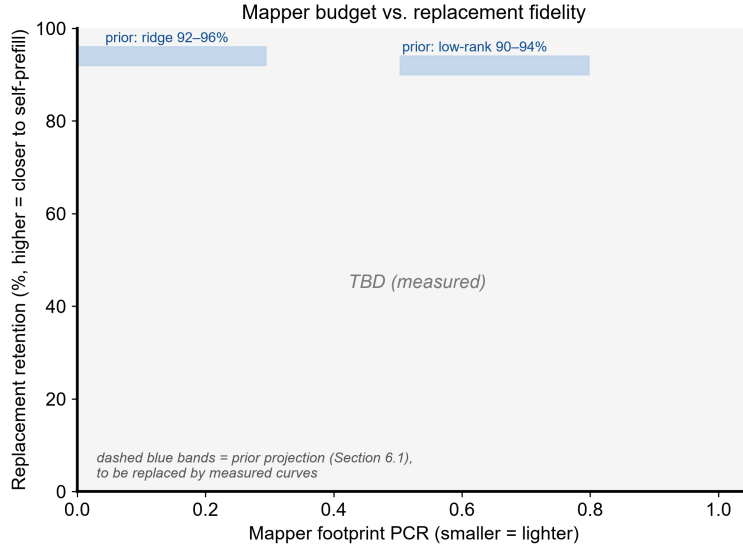


Figure 3: Mapper footprint (PCR) versus replacement retention. Dashed blue bands are prior projections from Section 6.1; the measured curve is pending.

PCR $\approx$ 0.23 is expected to lower retention only modestly, to roughly 90–94%. These bands, not single numbers, are the actual prediction: the published cross-model ridge results that motivate them report a range of pair-dependent fidelities, and there is no principled basis for asserting a single decimal before any Qwen3-4B→1.7B run has happened. If the measured curve falls inside this band, high-fidelity replacement and positive capability transfer should still be treated as distinct gates rather than inferring capability from KV or attention similarity.

6.3 Teacher Advantage Transfer

Under a wide but centered prior, full APCS plausibly yields a full-test CHG somewhere in the range of about –1.6 to +2.7 pp, corresponding to roughly –16% to +27% teacher-gap recovery, a band whose midpoint is only mildly positive and does not rule out $\text{CHG} \leq 0$. More importantly, the gain is projected to be heterogeneous with teacher gap: the plausible range for CHG should include zero when the teacher gap is small (below roughly 3–5 pp), and only shift confidently positive once the teacher exceeds the student by more than about 10 pp, with TGRR plausibly saturating somewhere between 10% and 25%. If observed, a widening gap between the CHG range and zero as teacher gap grows would be stronger evidence for runtime-transferable advantage than any single average accuracy delta; a flat, gap-independent CHG close to zero would instead support the direction-asymmetry concern raised in Section 7.

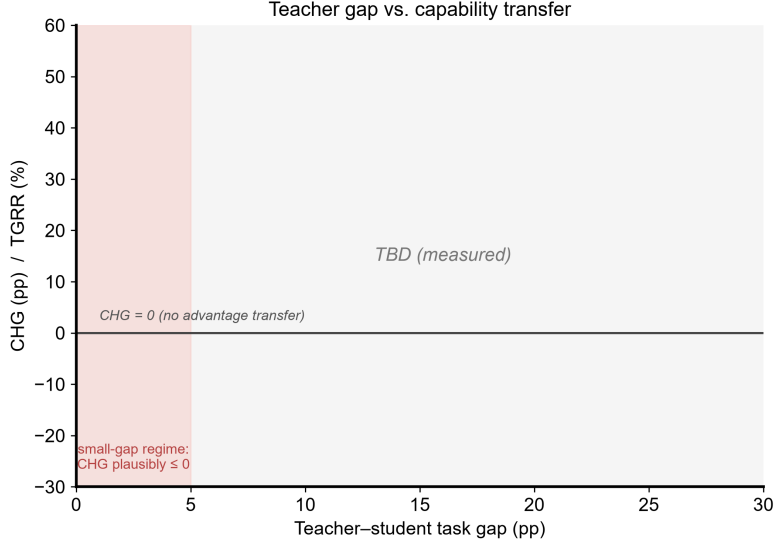


Figure 4: Teacher gap versus CHG/TGRR. The small-gap regime where CHG plausibly stays at or below zero is marked; measured points are pending.

6.4 Quality–Latency Pareto

The projected Pareto landscape places ridge and the low-rank base in the “prefill saving but below self-prefill quality” region with reasonable confidence, since both mappers are explicitly optimized toward the student’s own self-KV. APCS is different: its plausible range straddles $\text{CHG} = 0$, so whether it actually enters the paper’s target quadrant (positive CHG together with positive PSR, roughly 8–16% in the prior) is the open empirical question rather than a foregone conclusion. This is therefore a decisive figure precisely because its outcome is uncertain: if measured APCS lands at $\text{CHG} \leq 0$, the manuscript should pivot to Efficient State Handoff rather than continue to claim Runtime Capability Transfer; if it lands clearly above zero and grows with teacher gap, that is the paper’s central positive result.

6.5 Context-Length Scaling

The single-GPU cost prior predicts that map+load+query is slower than student full prefill at 1K ($\text{PSR} \approx -82\%$), and becomes progressively more favorable at longer contexts, but the exact break-even point depends on an assumption we cannot yet verify: whether map/load overhead is genuinely fixed or partly scales with context length. Under an optimistic fixed-overhead scenario the crossover falls around 4K ($\text{PSR} \approx +12\%$) and reaches roughly $+48\%$ by 16K; under a more conservative scenario where mapping cost itself grows with sequence length, the crossover shifts out to somewhere between 4K and 8K and the 16K figure drops to roughly $+30\%$. We report both scenarios rather than a single curve, and treat the true break-even as lying somewhere in the 4K–8K range rather than a fixed point. This distinction matters because the pilot’s single-24GB-GPU setup requires sequential teacher/student residency (load $\rightarrow$ capture $\rightarrow$ unload $\rightarrow$ load), which is a research-probe artifact rather than the production regime that the map+load+query cost model describes; the measured study must separately report model load/unload time, map time, cache H2D time, query-only prefill, and student full-prefill latency so the two are not conflated.

6.6 Behavioral Stability and Multi-turn Handoff

The multi-turn prior assumes gradual rather than catastrophic drift, but we now show it with a band that widens over turns rather than a single smooth line, since uncertainty about drift compounds the further we extrapolate from an untested turn-1 estimate: CHG plausibly falls from a range of about 0.2–2.2 pp at turn 1 to a range of about -2.1 – 2.3 pp at turn 20, decision consistency plausibly falls from roughly 97–98% toward a range as wide as 84–90% by turn 20, and next-token KL rises with similarly growing uncertainty.

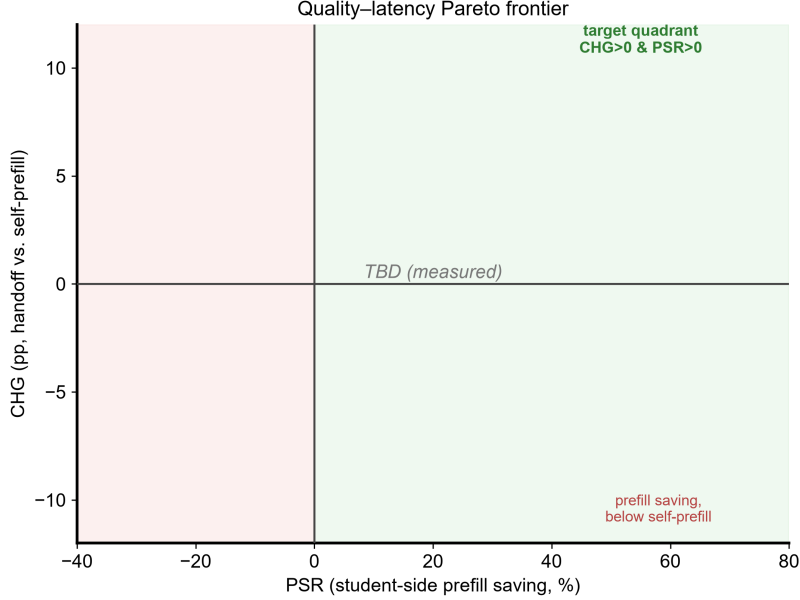


Figure 5: CHG/TGR-PSR Pareto frontier. The target quadrant ($\text{CHG} > 0$ and $\text{PSR} > 0$) is marked; measured points are pending.

Table 2: Core ablations, reported as plausible ranges. The Direct RoPE Mapping row uses a rounded band rather than a suspiciously exact figure. The final analysis should distinguish mechanisms that improve task-level handoff from those that only improve representation similarity.

Ablation	Retention	CHG	TGR	PSR	PCR	Interpretation
Full Ridge	92–96%	−4.4 to −2.2	−44 to −22%	3–8%	1.00	High fidelity, expensive; no advantage tra
Low-rank Base	90–94%	−5.5 to −3.3	−56 to −33%	12–20%	0.23	Compact, but remains emulation
+ Advantage State	98–104%	−1.0 to +2.0	−10 to +20%	10–16%	0.27	Initial positive capability signal
− Teacher Guidance	95–101%	−2.5 to +0.3	−25 to +3%	10–16%	0.27	CHG turns negative without teacher guid
− Self Stability	100–106%	−0.2 to +3.2	−2 to +32%	10–16%	0.27	Higher CHG but projected behavioral dr
Direct RoPE Mapping	79–90%	−11.5 to −5.5	−115 to −55%	8–14%	0.27	Position mismatch causes large fidelity lo
Full APCS	97–105%	−1.6 to +2.7	−16 to +27%	8–16%	0.27	Plausibly the best trade-off, but CHG ra

The widening bands are the point: we would not be surprised by a turn-20 CHG close to zero, and we do not treat the turn-1 central estimate as a reliable predictor of long-horizon behavior. If measured drift is materially faster than even the pessimistic edge of this band, periodic refresh or selective re-prefill should become a primary follow-up rather than reporting only first-turn quality.

7 Planned Ablations

The projected ablation hypothesis is that the compatibility base alone does not create positive CHG; an advantage state is necessary for the capability signal; removing teacher guidance pushes CHG back below zero; removing self-stability can increase short-term CHG at the cost of larger behavioral drift; and directly mapping RoPE-encoded keys causes the largest retention failure. Offline experiments should treat these as falsifiable predictions, not target numbers that must be reproduced.

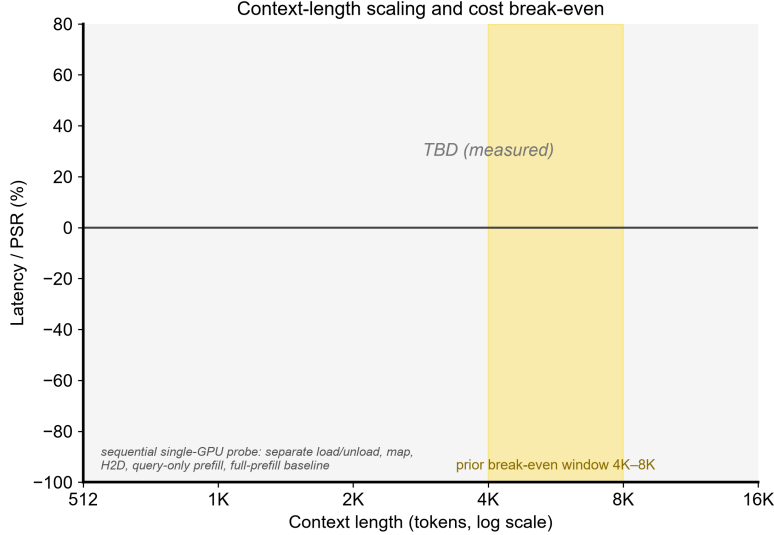


Figure 6: Context-length scaling and cost break-even. The 4K–8K window is a prior projection; measured latency decompositions are pending.

8 Discussion

The experiments are designed so that their interpretation is fixed in advance. If CHG is consistently positive, the analysis turns to where the advantage lives: whether TGRR shows reproducible structure across teacher gap, layer, head, task type, and context length. If CHG stays at or below zero while replacement retention is high, the conclusion is that state compatibility does not imply capability transfer.

8.1 Negative Results and Direction Asymmetry

Large-to-small transfer may be intrinsically more difficult than small-to-large transfer because the student can have insufficient representational capacity or incompatible attention geometry for teacher-specific state. If $\text{CHG} \leq 0$ while replacement retention remains high, an important conclusion is that cross-model state compatibility does not imply capability transfer. If capability transfer appears only for certain pairs, the representational geometry that predicts directional success becomes a mechanism result in its own right.

8.2 Systems and Security Boundaries

PSR claims apply primarily when the teacher cache already exists because of an upstream task. If the teacher is run solely to create transferable state, teacher prefill must be included and the reuse break-even point reported. Persisted KV state may encode sensitive information about the original context; productionization therefore requires cache lifecycle controls, access isolation, encryption at rest/in transit, minimization of persisted state, and verifiable deletion.

9 Conclusion

The question we set out to test is whether a weaker student can skip full prefill of the original context, consume only a stronger teacher’s runtime state, and inherit part of the teacher’s task advantage. The present draft contains no measured answer; it fixes the probe, the expectations, and the decision rule. The probe runs on a sub-12GB GPU, uses roughly 70 lines of code on top of Transformers, and completes in under half a day (Sections 5.5 and Appendix A). The final paper claims Runtime Capability Transfer only if frozen-test experiments establish $\text{CHG} > 0$ with a confidence interval excluding zero, consistently positive

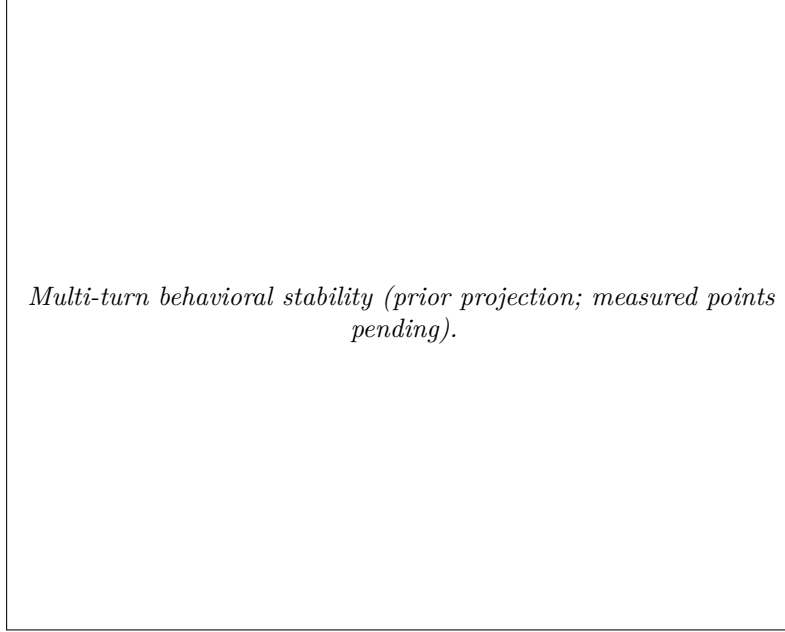


Figure 7: Multi-turn behavioral stability (prior projection; measured points pending).

TGRR, and $\text{PSR} > 0$. Otherwise it reports Efficient State Handoff, or the boundary result that state compatibility does not imply capability transfer.

Ethics, Security, and Reproducibility

Ethics and security. The research does not require collection of personal data. If real operational contexts are later used, data classification, de-identification, and KV-cache lifecycle controls are required. We will report negative results, all seeds, and full frozen test sets rather than selecting examples to manufacture positive CHG.

Reproducibility. The final artifact should pin model commits, tokenizer, dtype, attention implementation, RoPE configuration, cache layout, random seeds, hardware, and context lengths. Each run should preserve configuration, metrics, system measurements, and logs.

AI use statement (update before submission). Generative AI was used in preparing this manuscript draft for structural planning, language drafting, and formatting. The authors remain responsible for experimental decisions, code validation, data analysis, result interpretation, reference verification, and all final scientific claims. This statement must be updated to match actual usage and the ICLR policy in force at submission time.

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A Reference Implementation and Engineering Details

A.1 Core Implementation: Ridge Fit and 4-Way Intervention

All four interventions are operations on the HuggingFace `past_key_values` tuple. Step 1 extracts teacher and student KV on a 100-sample calibration set, de-RoPEs keys before fitting, and computes per-layer, per-head ridge mappings W_K and W_V with the closed-form solution. Step 2 assembles the four KV combinations and evaluates perplexity on continuation tokens.

1. Calibration capture and ridge fit.

```
# capture teacher KV (source state) and student KV (reference target)
def capture_kv(model, input_ids):
    out = model(input_ids, use_cache=True, output_hidden_states=False)
    return out.past_key_values          # tuple of (K, V) per layer

# de-RoPE keys before fitting; re-RoPE after mapping (Section A.2.1)
```

```

def derope_k(k, positions, inv=False):
    # k: [B, H, T, D] with D even; RoPE base shared by teacher and student
    D = k.shape[-1]
    theta = positions.float().unsqueeze(-1) / model.ropes_base ** (
        torch.arange(0, D, 2, device=k.device).float() * 2 / D)
    cos, sin = theta.cos(), theta.sin()
    k1, k2 = k[..., 0::2], k[..., 1::2]
    if inv: k1, k2 = k2, k1 # inverse rotation (de-RoPE)
    return torch.cat([k1 * cos - k2 * sin, k1 * sin + k2 * cos], dim=-1)

def fit_ridge(Z, Y, lam=1e-3): # per-layer, per-head [B,H,T,D]
    Zt = Z.transpose(-1, -2)
    return torch.linalg.solve(Zt @ Z + lam * torch.eye(D, device=Z.device), Zt @ Y)

# fit on the calibration set: W maps teacher KV to student KV
W_k, W_v = {}, {}
for l in range(student_layers): # 28 student layers
    tl = int(round(l * (36 - 1) / (28 - 1))) # layer alignment (A.2.2)
    W_k[l] = fit_ridge(Z_k_teacher[tl].flatten(0, 2), K_student[l].flatten(0, 2))
    W_v[l] = fit_ridge(Z_v_teacher[tl].flatten(0, 2), V_student[l].flatten(0, 2))

2. Four-way intervention PPL evaluation loop.

def evaluate_ppl(student_model, input_ids, custom_kv=None):
    with torch.no_grad():
        outputs = student_model(input_ids=input_ids, past_key_values=custom_kv,
                                labels=input_ids)
        return torch.exp(outputs.loss).item()

def mapped_kv(teacher_kv, W_k, W_v, positions):
    kv = []
    for l in range(student_layers):
        tl = int(round(l * 35 / 27))
        k = teacher_kv[tl][0] # teacher key, RoPE-encoded
        k = derope_k(k, positions, inv=True) # de-RoPE content space
        k = k @ W_k[l] # ridge map to student
        k = derope_k(k, positions, inv=False) # re-RoPE with student scheme
        kv.append((k, teacher_kv[tl][1] @ W_v[l])) # mapped K and V
    return tuple(kv)

# the 4 intervention arms switch only this tuple:
kv_case1 = (native_k, native_v) # Native K + Native V (Baseline)
kv_case2 = (mapped_k, native_v) # Mapped K + Native V
kv_case3 = (native_k, mapped_v) # Native K + Mapped V
kv_case4 = (mapped_k, mapped_v) # Mapped K + Mapped V (Full Handoff)
ppl = [evaluate_ppl(student, ids, kv)
        for kv in [kv_case1, kv_case2, kv_case3, kv_case4]]

```

Padding handling. PPL is computed only on target (continuation) tokens; padding tokens are excluded by setting their label to `-100` (`ignore_index`) rather than by removing them from the sequence, so the attention and cache layout stays unchanged.

A.2 Critical Engineering Details (Three Pitfalls)

These details are not cosmetic. A nominal 512/1024-token experiment silently shrank to 70 tokens in an earlier pipeline because of global cropping, and a value diagnostic accidentally used values as both keys and values. The three rules below prevent that class of error.

1. RoPE alignment. Keys must be de-RoPE'd before ridge mapping and re-RoPE'd after mapping (Eq. 7); values are mapped in raw form. The teacher and student must share the same RoPE base. Within

Qwen3 the head dimension is 128 in both models, so the rotation can be inverted channel-wise and no head re-indexing is required.

2. Layer alignment (36 $\rightarrow$ 28). We use a fixed 1-to-1 correspondence via uniform linear interpolation, $\text{teacher_layer_idx} = \text{round}(\text{student_layer_idx} \times (36-1)/(28-1))$, and do not spend compute on learned layer-mixture weights in the first version.

3. Context-length bucketing and padding exclusion. Test texts are strictly filtered or truncated to $\text{max_length} \in \{512, 1024, 2048\}$ before evaluation, and PPL must exclude padding tokens from the loss (labels set to -100).

B Result Completion Checklist and Gated Workflow

- Main table with at least two large-to-small pairs and full-test Student/Teacher/Ridge/APCS results.
- Figure 3: PCR–Retention, establishing the mapper-budget/compatibility trade-off.
- Figure 4: Teacher Gap–CHG/TGRR, supporting or falsifying runtime-advantage transfer.
- Figure 5: CHG/TGRR–PSR Pareto frontier as the primary paper figure.
- Figure 6: context-length scaling with absolute latency decomposition.
- Figure 7: multi-turn behavioral stability with KL/JCR/consistency.
- Ablations on rank/shared basis, advantage state, teacher guidance, self stability, de-RoPE, and top- k layers.
- The T00–T11 gated workflow: T00 Compatibility Scanner $\rightarrow$ T01 Self-KV Replay $\rightarrow$ T02 RoPE $\rightarrow$ T03 Layer Alignment $\rightarrow$ T04 Ridge $\rightarrow$ T05 Replacement $\rightarrow$ T06 Lightweight Mapper $\rightarrow$ T07 Teacher Gap $\rightarrow$ T08 Advantage State $\rightarrow$ T09 CHG/TGRR $\rightarrow$ T10 PSR $\rightarrow$ T11 MVP Decision. A failed gate blocks downstream claims so that every paper result is traceable to a verified engineering stage.